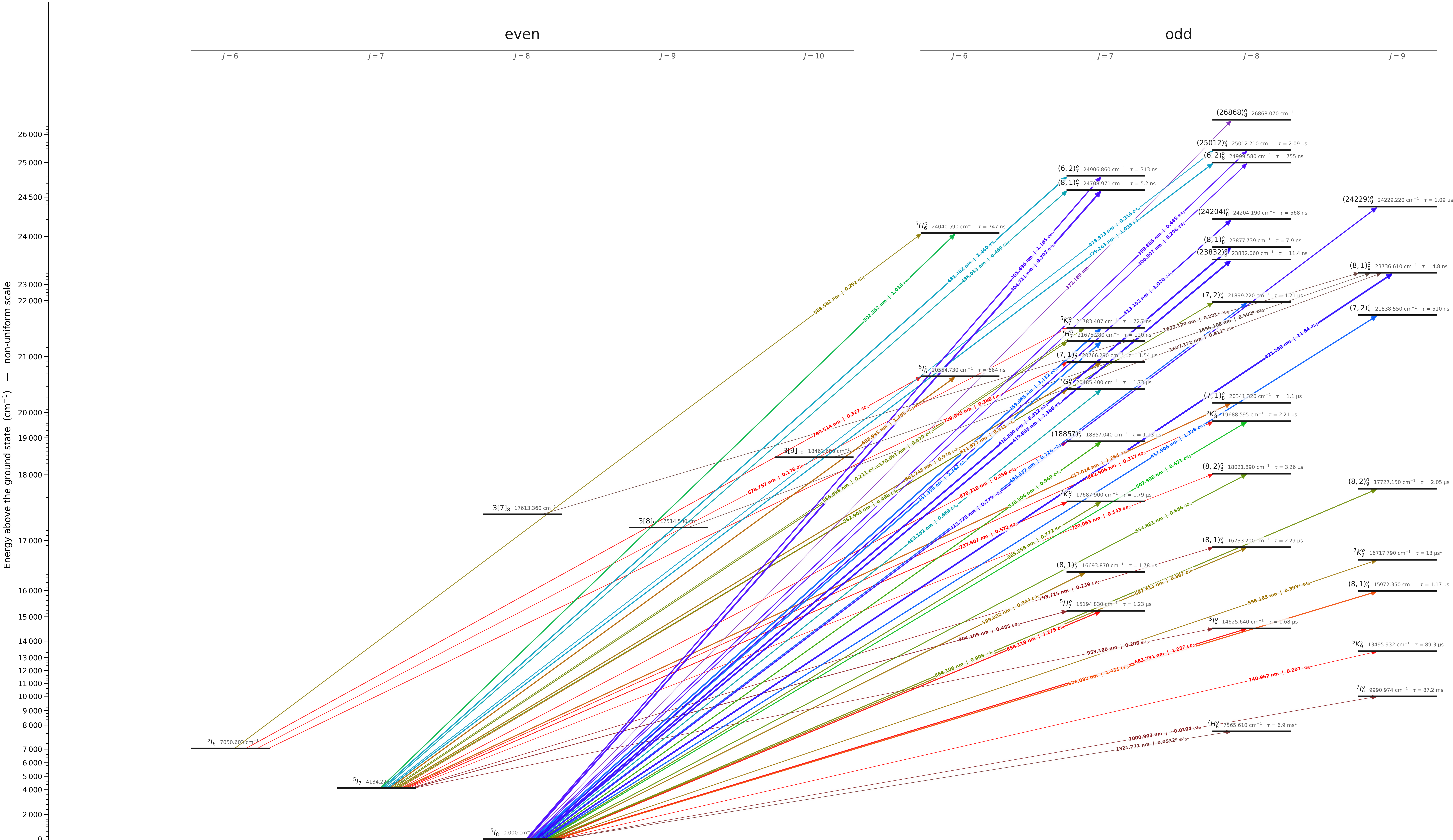
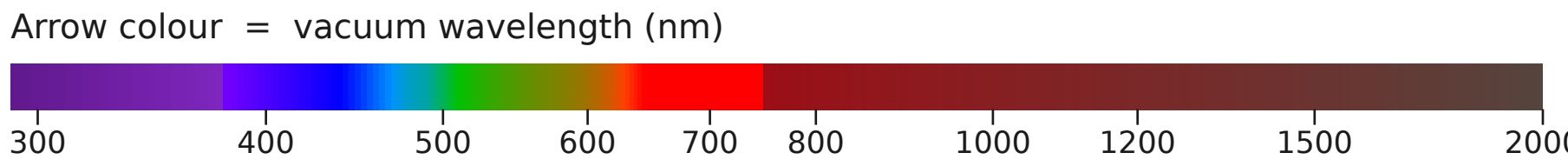


Dysprosium-164

¹⁶⁴Dy (Dy II), nuclear spin *I* = 0. Energy levels and transitions below 2 μm: 39 levels, 55 lines.

Dy⁺ ionisation limit 47901.76 cm⁻¹ = 5.93906 eV (208.761 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U||\langle r||J\rangle|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	U er J' (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
5f ₈ - (26868)g	372.1890	372.0832	805.4845	-	-	-	-	UV ground-state line
5f ₈ - (8, 1)g	404.7113	404.5970	740.7563	9.707	1.920e+08	30.61	99.8	strong blue J=8->7 line
5f ₈ - (8, 1)g	418.8001	418.6821	715.8366	8.812	1.260e+08	20.15	99.5	strong blue J=8->8 line
5f ₈ - (23832)g	419.6028	419.4846	714.4672	7.386	8.800e+07	13.96	100	strong blue J=8->8 line
5f ₈ - (8, 1)g	421.2902	421.1715	711.6057	11.84	2.000e+08	33.16	100	main broad cooling line: Zeeman slower, transverse coolin...
5f ₈ - (7, 2)g	456.6373	456.5093	656.5221	0.726	6.600e+05	0.1315	79.9	pure 6s->6p reference line for King plots
5f ₈ - (8, 2)g	564.1065	563.9499	531.4466	0.908	4.900e+05	0.07764	100	closed J=8->9 narrow line
5f ₈ - 7Kg	598.1652	597.9995	501.1867	0.393*	7.700e+04	0.01224	100	"598 nm" closed J=8->9 narrow line proposed for narrow-li...
5f ₈ - (8, 1)g	599.0223	598.8564	500.4696	0.944	5.600e+05	0.08941	99.7	narrow J=8->7 probe line
5f ₈ - (8, 1)g	626.0819	625.9088	478.8390	1.431	8.900e+05	0.136	100	narrow-line MOT
5f ₈ - 5Hg	658.1186	657.9368	455.5295	1.275	7.700e+05	0.1294	94.7	narrow probe line used for Doppler thermometry of the 421...
5f ₈ - 5f _g	683.7308	683.5421	438.4657	1.257	5.893e+05	0.09474	99	narrow J=8->8 optical pumping / demagnetization cooling l...
5f ₈ - 5Kg	740.9619	740.7579	404.59900(10)	0.207	1.120e+04	0.001782	100	ultra-narrow-line MOT / narrow-line cooling in tweezers
5f ₈ - 7f _g	1000.9034	1000.6291	299.5219	0.010	1.150e+01	1.825e-06	100	ultra-narrow
5f ₈ - 7Hg	1321.7705	1321.4091	226.8113	0.053*	1.460e+02	2.307e-05	100	proposed ultra-narrow telecom-band line

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
5f ₈ - (26868)g	372.189	164-163	-44(10)
5f ₈ - (26868)g	372.189	164-162	-178(13)
5f ₈ - (26868)g	372.189	164-161	-194(13)
5f ₈ - (26868)g	372.189	164-160	-339(13)
5f ₈ - (8, 1)g	421.290	164-163	-616.3(5)
5f ₈ - (8, 1)g	421.290	164-162	-913.2(8)
5f ₈ - (8, 1)g	421.290	164-161	-1635.0(10)
5f ₈ - (8, 1)g	421.290	164-160	-1895.0(20)
5f ₈ - (8, 1)g	421.290	164-158	-2868(9)
5f ₈ - (8, 1)g	421.290	164-156	-4300(15)
5f ₈ - (7, 2)g	456.637	164-163	-660(3)
5f ₈ - (7, 2)g	456.637	164-162	-971.0(20)
5f ₈ - (7, 2)g	456.637	164-161	-1744(3)
5f ₈ - (7, 2)g	456.637	164-160	-2020(3)
5f ₈ - (7, 2)g	456.637	164-158	-3061(4)
5f ₈ - (7, 2)g	456.637	164-156	-4604(5)
5f ₈ - (8, 2)g	564.106	164-163	-685.0(20)
5f ₈ - (8, 2)g	564.106	163-162	-328(3)
5f ₈ - (8, 2)g	564.106	162-161	-801(3)
5f ₈ - (8, 2)g	564.106	161-160	-284.0(20)
5f ₈ - (8, 2)g	564.106	164-162	-1012.0(20)
5f ₈ - (8, 2)g	564.106	162-160	-1086(5)

Shift = ν(first isotope) - ν(second isotope).